Grundlagen "Web-Engineering"

World Wide Web

"HyperText is a way to link and access information of various kinds as a web of nodes in which the user can browse at will. Potentially, HyperText provides a single user-interface to many large classes of stored information such as reports, notes, data-bases, computer documentation and on-line systems help. We propose the implementation of a simple scheme to incorporate several different servers of machine-stored information already available at CERN, including an analysis of the requirements for information access needs by experiments."

(https://www.w3.org/Proposal.html)

Das **World Wide Web** (englisch für „weltweites Netz“, kurz Web oder WWW) ist ein über das **Internet** abrufbares System von [...] sogenannten **Webseiten**, welche mit **HTML** beschrieben werden.

Sie sind durch **Hyperlinks** untereinander verknüpft und werden im Internet über die Protokolle **HTTP oder HTTPS** übertragen.

Zum Aufrufen von Inhalten aus dem World Wide Web wird ein **Webbrowser** benötigt, der z. B. auf einem **PC** oder einem **Smartphone** läuft. Mit ihm kann der Benutzer die auf einem beliebigen, von ihm ausgewählten **Webserver** bereitgestellten Daten herunterladen und auf einem geeigneten Ausgabegerät wie einem **Bildschirm** oder einer **Braillezeile** anzeigen lassen.

(vgl. https://de.wikipedia.org/wiki/World_Wide_Web)

Engineering

Engineering is the practice of using **natural science**, **mathematics**, and the **engineering design** process to solve **technical problems**, increase **efficiency** and **productivity**, and **improve systems**.

The **engineering design process** is a common series of steps that engineers use in creating **functional products and processes**. The process is **highly iterative**[...]

It is a **decision making process** in which the **basic sciences** [...] are **applied** to convert resources optimally **to meet a stated objective**.

Among the fundamental elements of the design process are the **establishment of objectives and criteria**, **synthesis**, **analysis**, **construction**, **testing** and **evaluation**.

(vgl. https://en.wikipedia.org/wiki/Engineering, https://en.wikipedia.org/wiki/Engineering_design_process)

- **Anwendung von Wissenschaften**
- Lösen von technischen Problemen, erhöhen der Effizienz, Optimierung von Systeme
- **Iterativer Prozess**
- Entscheidungsverfahren
- Bestimmen der Anforderungen und **Analyse**, **Herstellung**, **Überprüfung** und **Evaluation**

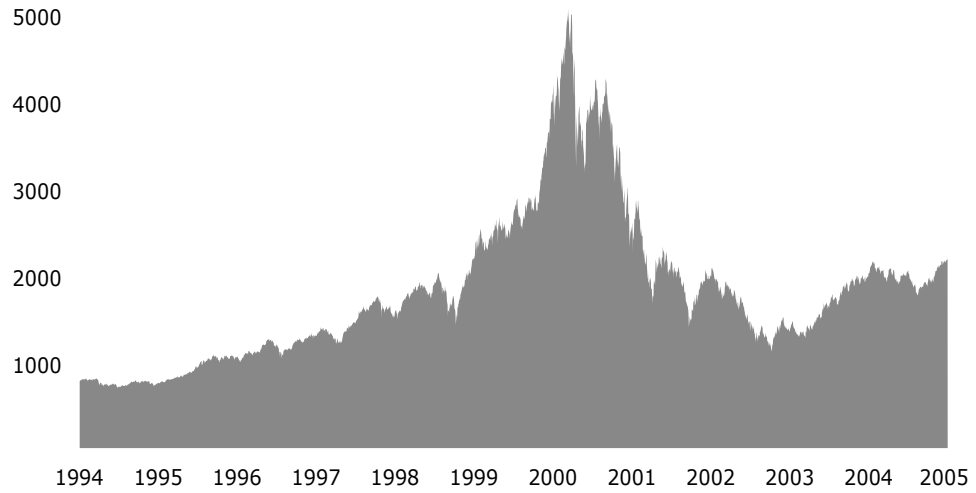
History of the Web

Web 1.0

- WWW (1989): Tim Berners-Lee, CERN: https://info.cern.ch/hypertext/WWW/TheProject.html
- HTTP (1991): https://datatracker.ietf.org/doc/html/rfc1945
- W3C (1994): https://www.w3.org/

(vgl. https://en.wikipedia.org/wiki/History_of_the_World_Wide_Web)

dot-com bubble



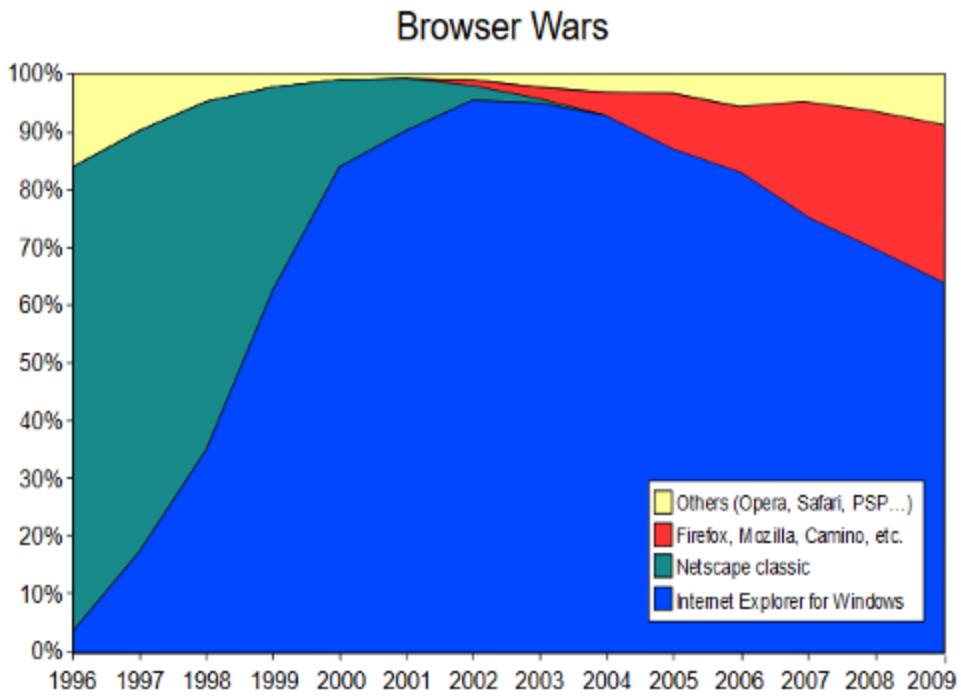
Nasdaq Composite index

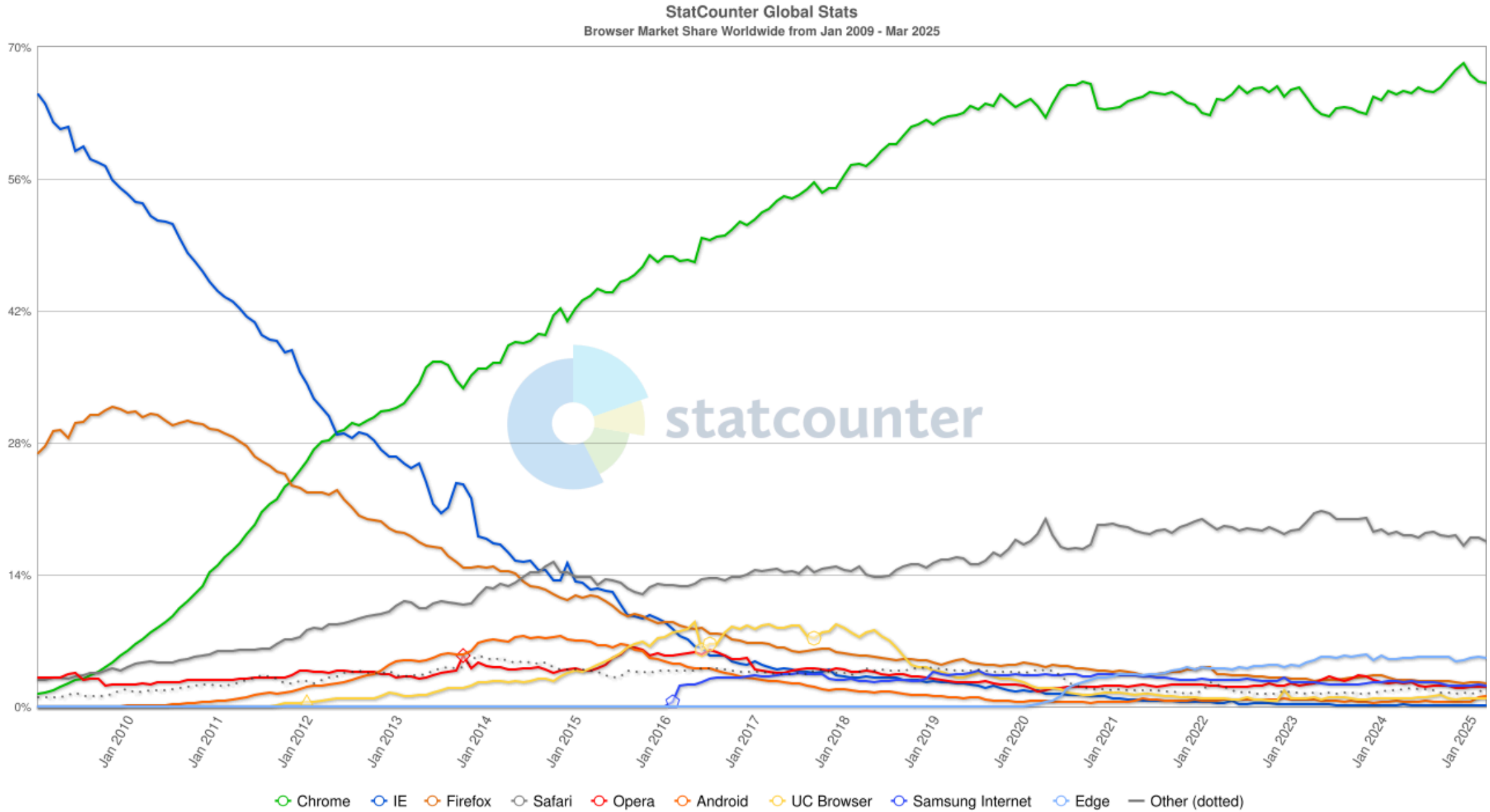
https://en.wikipedia.org/wiki/Dot-com_bubble

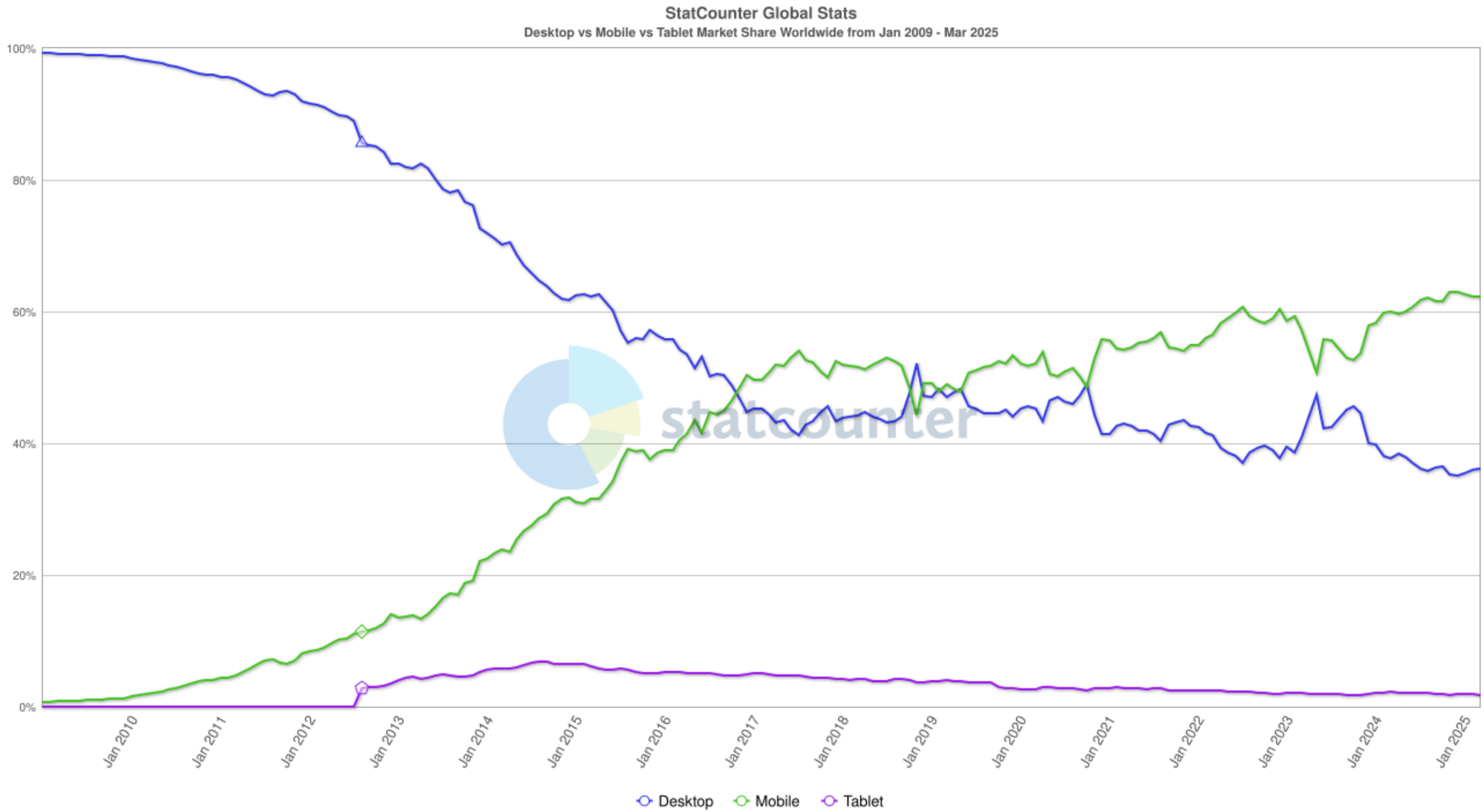
Web 2.0

- Social Media: Youtube (2005), Facebook (2004), MySpace (2003), Twitter (2006)
- JavaScript (1995): Brendan Eich, Netscape
- Ajax (2000): XMLHttpRequest
- Web Applications, SPA, PWA: Gmail (2004), Google Maps (2005)
- Cloud Computing: AWS (2002)
- Mobile: iPhone (2007)

Browser Wars




StatCounter Global Stats
Browser Market Share Worldwide from Jan 2009 - Mar 2025
70%
56%
42%
28%
14%
0%
statcounter
Jan 2010
Jan 2011
Jan 2012
Jan 2013
Jan 2014
Jan 2015
Jan 2016
Jan 2017
Jan 2018
Jan 2019
Jan 2020
Jan 2021
Jan 2022
Jan 2023
Jan 2024
Jan 2025
Chrome
IE
Firefox
Safari
Opera
Android
UC Browser
Samsung Internet
Edge
Other (dotted)


StatCounter Global Stats
Desktop vs Mobile vs Tablet Market Share Worldwide from Jan 2009 - Mar 2025
100%
80%
60%
40%
20%
0%
statcounter
Jan 2010
Jan 2011
Jan 2012
Jan 2013
Jan 2014
Jan 2015
Jan 2016
Jan 2017
Jan 2018
Jan 2019
Jan 2020
Jan 2021
Jan 2022
Jan 2023
Jan 2024
Jan 2025
Desktop
Mobile
Tablet

Web 3.0 / Semantic Web

- Tim Berners-Lee, 1999
- Daten werden mit Metadaten maschinenlesbar aufbereitet
- Dadurch können Informationen verlinkt werden
- Beispiel: RSS, https://hnrss.org/frontpage

Web3

- Dezentrales Web
- Blockchains
- smart contracts
- digital tokens

Gegenwart und Zukunft

- Browser als App-Plattform
- Local First: https://localfirstweb.dev/
- Internet of Things?
- "Dead Internet Theory" / Enshittification / AI Slop
  - https://www.wired.com/story/tiktok-platforms-cory-doctorow/
  - https://www.zeit.de/digital/internet/2025-04/plattformverfall-enshittification-cory-doctorow-facebook-internet

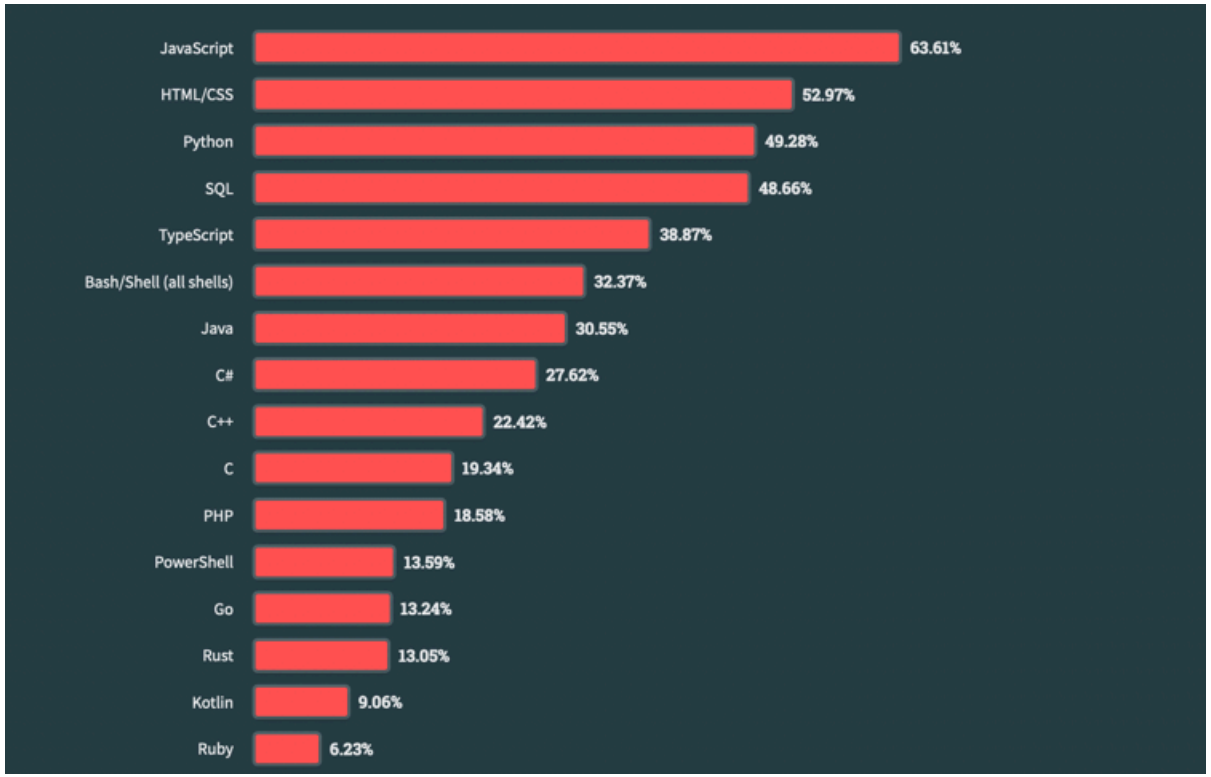
Werkzeuge

- https://developer.mozilla.org/
- https://www.mozilla.org/de/firefox/windows/
- https://www.google.com/intl/de/chrome/
- https://www.jetbrains.com/webstorm/
- Git: https://git-scm.com/
- Node.js: https://nodejs.org/en
- Docker: https://www.docker.com/
- https://caniuse.com/

Tools.md

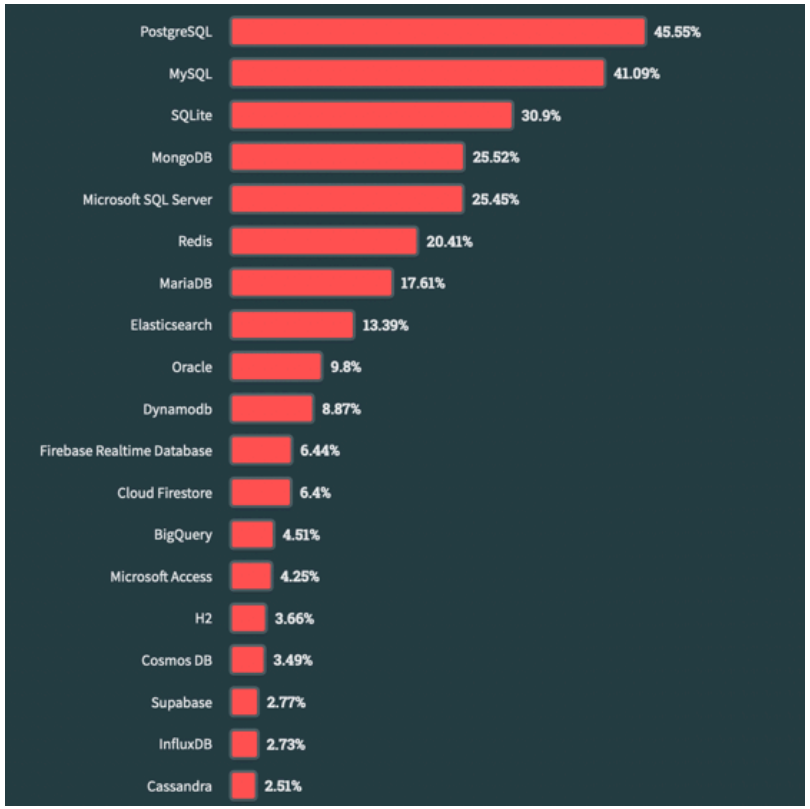
State of JavaScript

Programmiersprachen



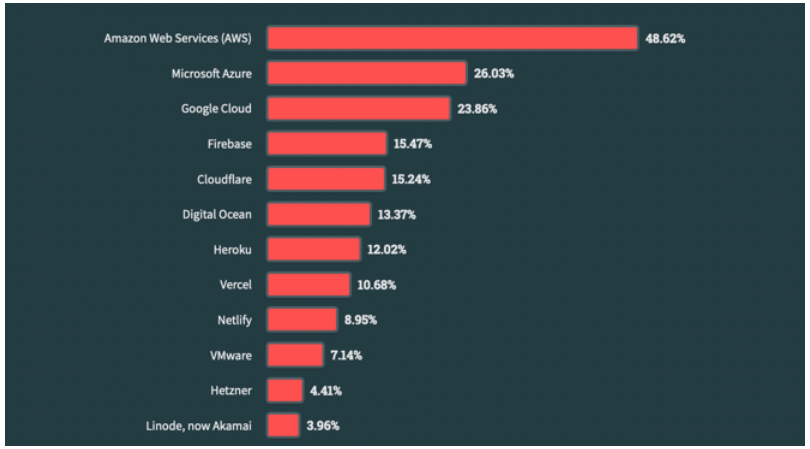
https://survey.stackoverflow.co/2023/#section-most-popular-technologies-programming-scripting-and-markup-languages,
23.04.24

Datenbanken



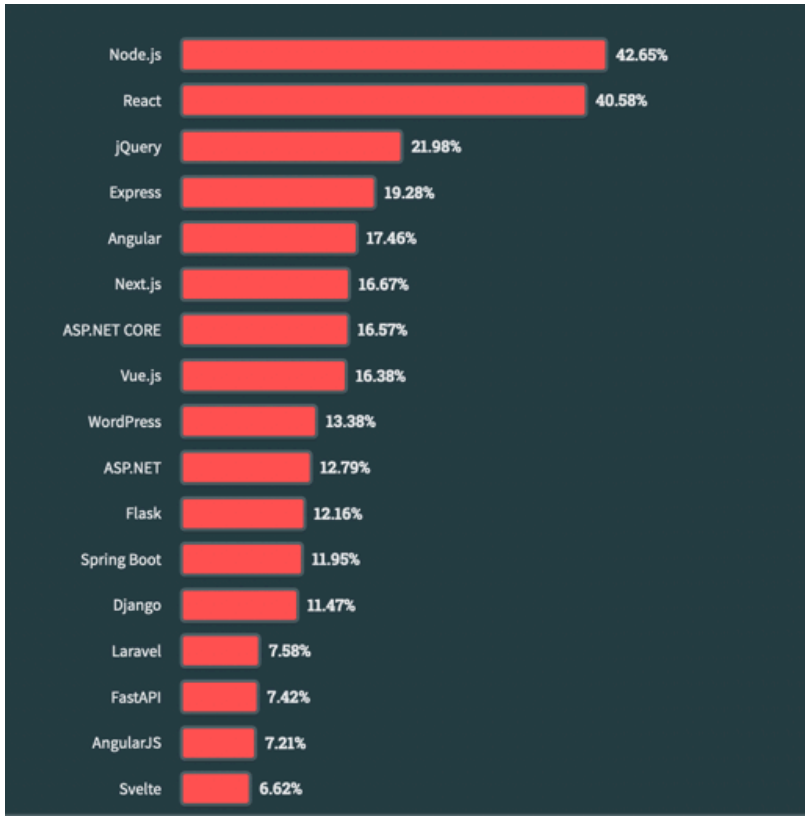
https://survey.stackoverflow.co/2023/#section-most-popular-technologies-databases, 23.04.24

Cloud Plattformen

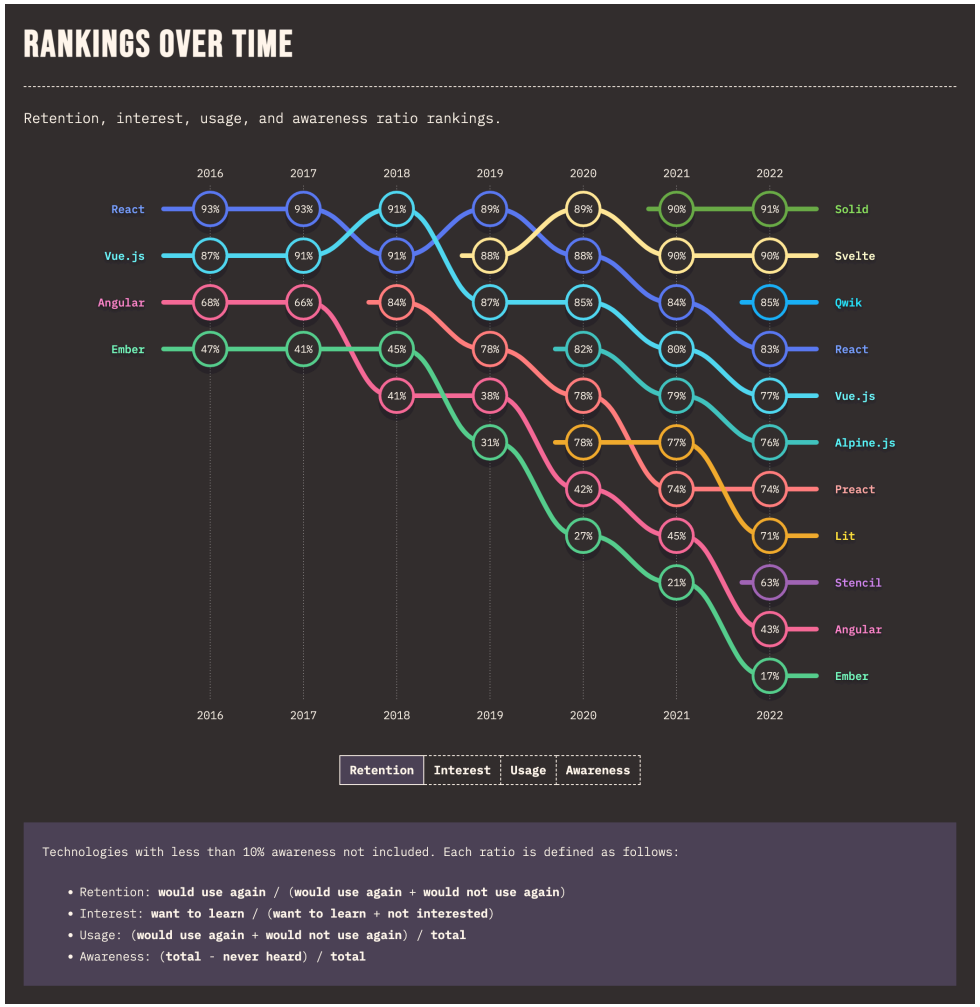


https://survey.stackoverflow.co/2023/#section-most-popular-technologies-cloud-platforms, 23.04.24

Web Frameworks



https://survey.stackoverflow.co/2023/#section-most-popular-technologies-web-frameworks-and-technologies, 23.04.24



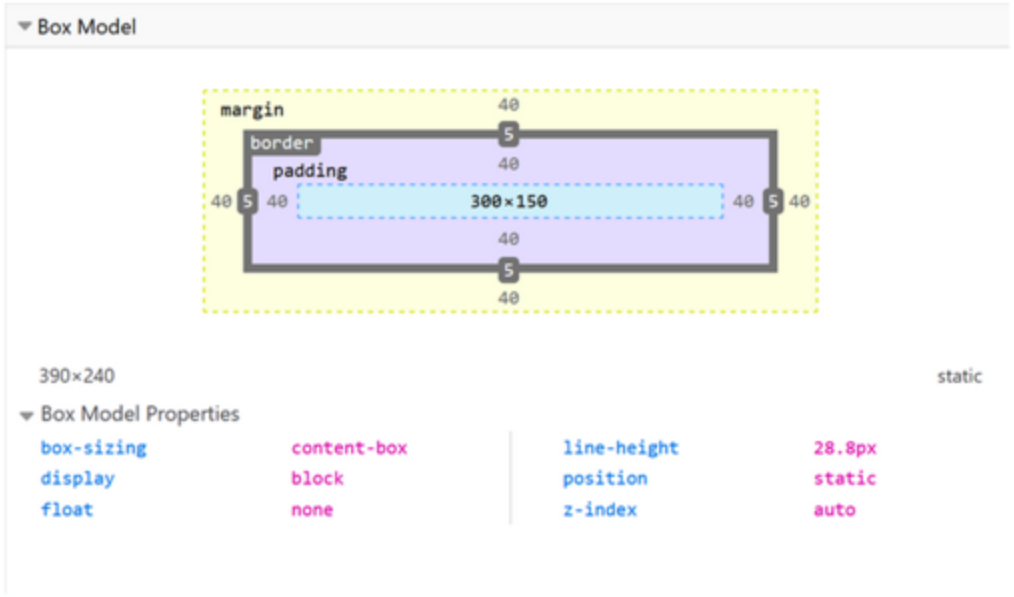
State of JavaScript 2022, 21.06.2023

Webseiten gestalten mit CSS

Layoutkonzepte

- http://info.cern.ch/hypertext/WWW/TheProject.html
- Framesets
- Tabellen
- Cascading Style Sheets (CSS)
- Fixed vs. Liquid Layout
- Responsive Webdesign
- Device Agnostic
- Mobile First

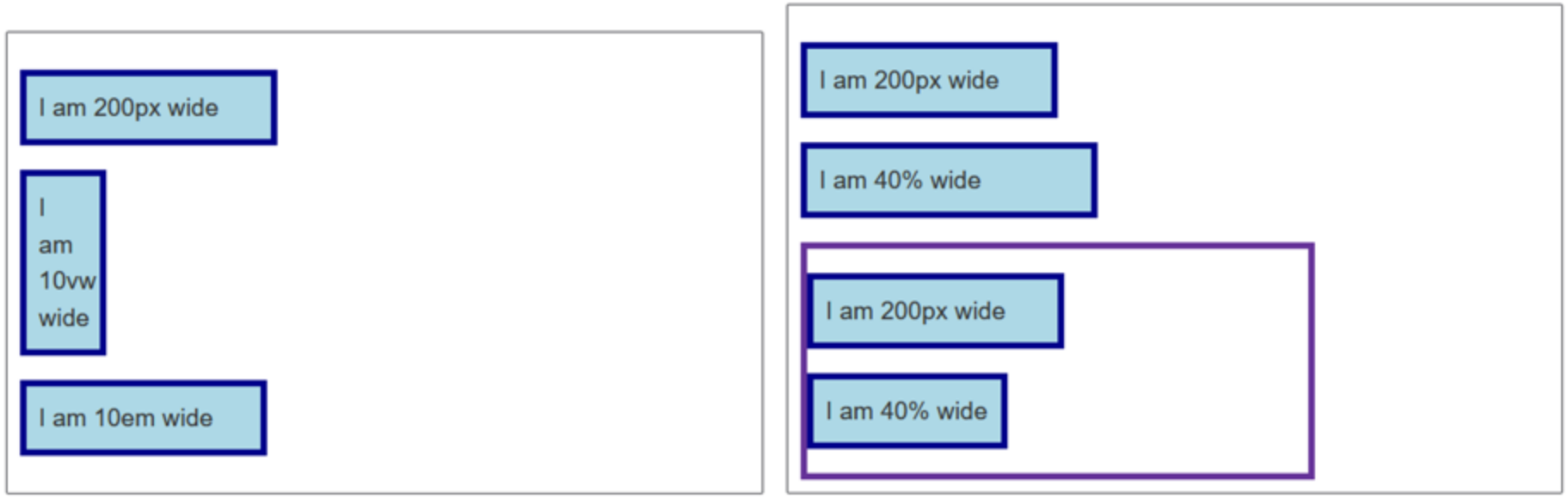
Box Model



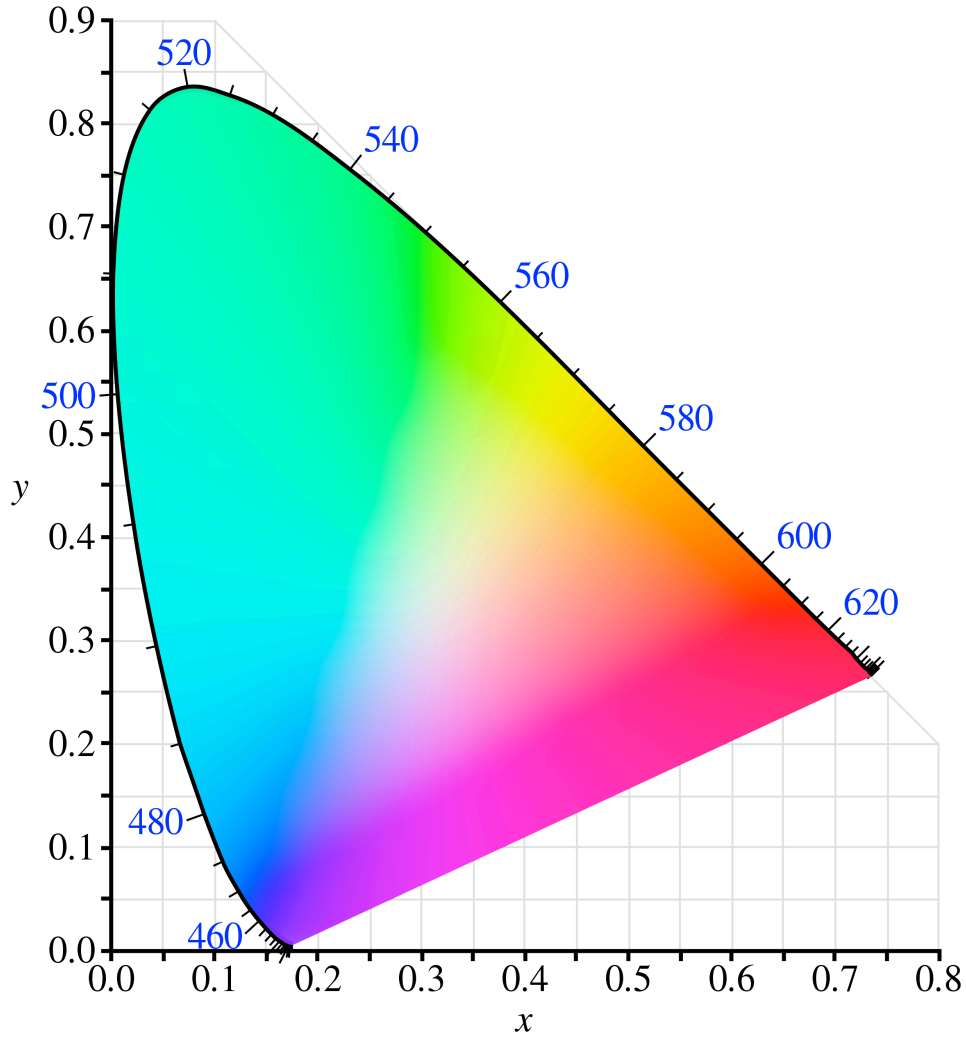
- `box-sizing: content-box` : `width` bezieht sich nur auf den content (blau, `300x150`)
- `box-sizing: border-box` : `width` bezieht sich auf content + padding + border (blau, violett, grau, `300+2*40+2*5` für die Breite)

Einheiten

- Absolute Grössen: `px` (`cm`, `mm`, ...) -> sparsam verwenden
- Relative Grössen
  - `em`: Schriftgrösse des Elternelements
  - `rem`: Schriftgrösse des Wurzelelements
  - `vw`, `vh`: viewport breite, viewport höhe

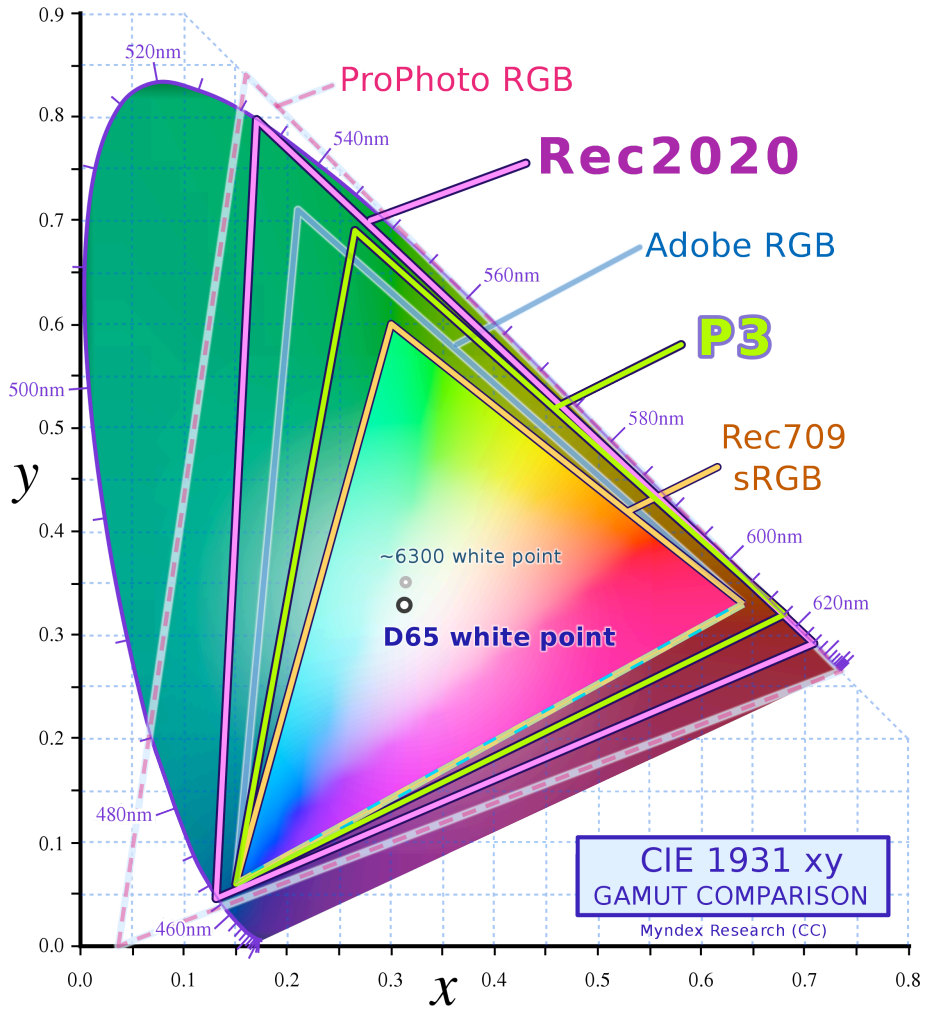


Farben



CIE 1931 Farbraum

Vergleich Farbräume



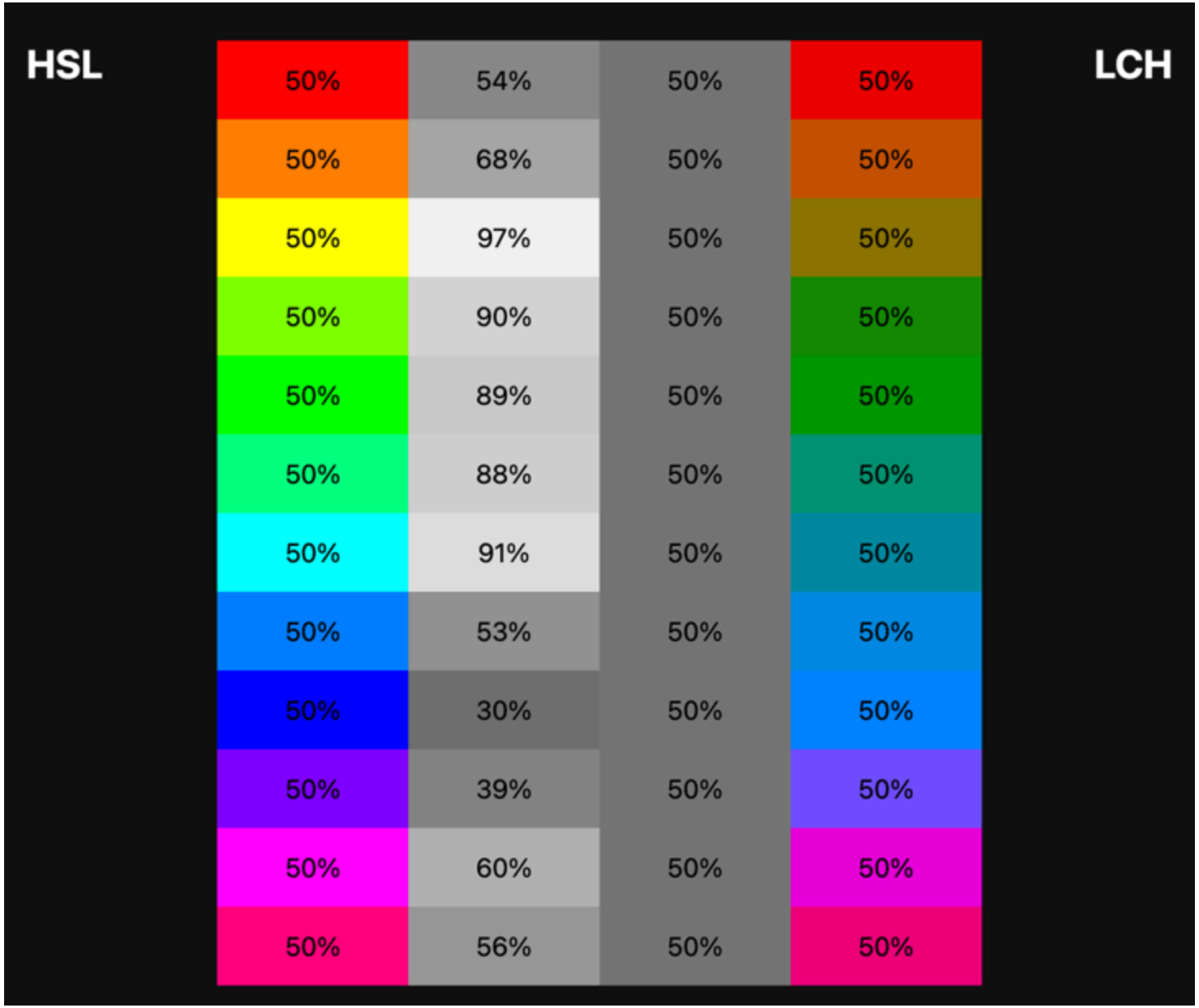
Farben in CSS

sRGB Farbraum

- Farbnamen: `color: darkblue;`
- Hex-Werte: `color: #ffa500;`
- RGBA-Werte (mit Deckkraft): `color: rgba(169, 169, 169, 0.5)`
- HSL-Werte (Hue, Saturation, Lightness): `color: hsl(60, 100, 50)`

Alle sichtbaren Farben

- LCH (Lightness Chroma Hue / Opacity): `color: lch(29.2345% 44.2 27 / 0.5)`
- Oklch: `color: oklch(40.1% 0.123 21.57)`
- CIELAB (Lightness, red-green, blue-yellow): `color: lab(29.2345% 39.3825 20.0664);`
- Oklab: `color: oklab(40.1% 0.1143 0.045);`



https://codepen.io/web-dot-dev/pen/peZqYxx

Webseiten interaktiv machen mit JavaScript

vgl.: Douglas Crockford (2018): How JavaScript Works, virgule solidus

How Class Free Works

- Klassen sind syntaktischer Zucker, d.h. sie bieten keine Funktionalität, die nicht mit anderen Mitteln erreicht werden
  kann.
- Sie verhalten sich anders als Klassen in C++, Java oder C#. Das kann verwirrend sein.

"Composition over Inheritance"

- Vererbung ist weniger zentral als manche Sprachen oder Kurse vermitteln.
- Vererbung bringt auch einige Probleme mit sich, da die Klassen sehr eng gekoppelt sind und nicht explizit klar ist,
  welche Methoden aufgerufen werden.
- Komposition ist sehr leistungsfähig.

Closures

Verschachtelte Funktionen können auf Variablen aus den äusseren Funktionen zugreifen.

Auch nach deren Ausführung.

```
function init() {
    var name = "Mozilla"; // name is a local variable created by init
    function displayName() {
        // displayName() is the inner function, that forms the closure
        console.log(name); // use variable declared in the parent function
    }

    displayName();
}

init();
```

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Closures

Folgende Struktur wird empfohlen:

```
function counter_constructor() {
    // private property
    let counter = 0;

    // composition
    const reuse = other_constructor();

    function up() {
        counter -= 1;
        return counter;
    }

    function down() {
        counter += 1;
        return counter;
    }

    // freeze to make the object immutable
    return Object.freeze({
        // make functions up and down public
        up,
        down,
        // expose goodness property from another object
        goodness: reuse.goodness
    })
}
```

Asynchronität

- JavaScript wurde primär für User-Interaktionen entwickelt.
- Asynchronität ist deshalb ein zentrales Sprachfeature.
- Es gibt verschiedene Möglichkeiten für asynchronen Code:
  - Callbacks
  - Promise
  - `async` / `await`

`async` / `await` ist verwirrend, weil damit Code produziert wird, der synchron aussieht, aber asynchron funktioniert.

Callback-Funktionen

- Callback-Funktionen werden als Parameter einer Funktion übergeben und von dieser aufgerufen.
- Die sogenannt "Callback-Hell", gemeint ist die Verschachtelung von Callbacks in Callbacks, sollte vermieden werden.

```
function foo(callback) {
    // some functionality

    callback(value);
}

foo((value) => {
    // runs after "some functionality"
})
```

Promise

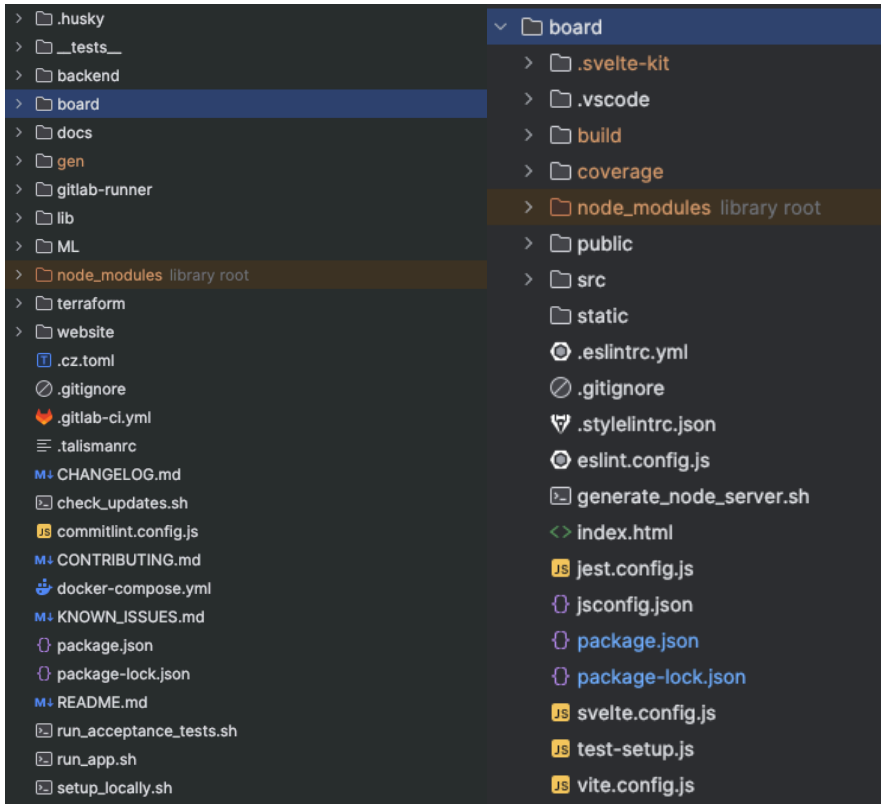
Promises können klarer sein als Callbacks, sind aber auch weniger explizit und potenziell verwirrend.

```
const p1 = new Promise((resolve, reject) => {
    // some functionality

    resolve("Success!");
});
p1.then((value) => {
        // runs after "some functionality"
    }
);
```

Single-Page-Applikationen implementieren

Typische Dokumentstruktur



- **Docs:** README.md, CONTRIBUTING.md, LICENCE.md, etc
- **Configs:** `.gitignore`, `.talismanrc`, `commitlint.config.js`, `eslint.config.js`, `.dockerignore`, `jest.config.js`, `jsconfig.json`,
- **Build & Deployment:** `.gitlab-ci.yml`, `Dockerfile`, `docker-compose.yml`, `package.json`, `package-lock.json`
- **Render & build outputs:** `/dist`, `/build`, `/coverage`, `/node_modules`
- **Source Code:** `/src`

Local First

1. No spinners: your work at your fingertips
2. Your work is not trapped on one device
3. The network is optional
4. Seamless collaboration with your colleagues
5. The Long Now
6. Security and privacy by default
7. You retain ultimate ownership and control

- https://www.inkandswitch.com/local-first/
- https://localfirstweb.dev/

Webservices implementieren

- Level 0: The Swamp of POX
- Level 1: Resources
- Level 2: HTTP verbs

URI	HTTP verb	Operation
/destinations	GET	retrieve destinations
/reservations	GET	get reservations according to certain criteria
/reservations	POST	add/cancel reservations
/rooms	POST	request room extras
/rooms	DELETE	remove room extras

- Level 3: Hypermedia controls

```
{
  customerId: "1",
  reservations: [
```

Die Performance von Webanwendungen optimieren

- Nachhaltige Software wird immer wichtiger
- Wirth's law:

> "The hope is that the progress in hardware will cure all software ills. However, a critical observer may observe that software manages to outgrow hardware in size and sluggishness."

- Wettbewerbsvorteil
- https://www.blauer-engel.de/de/produktwelt/software

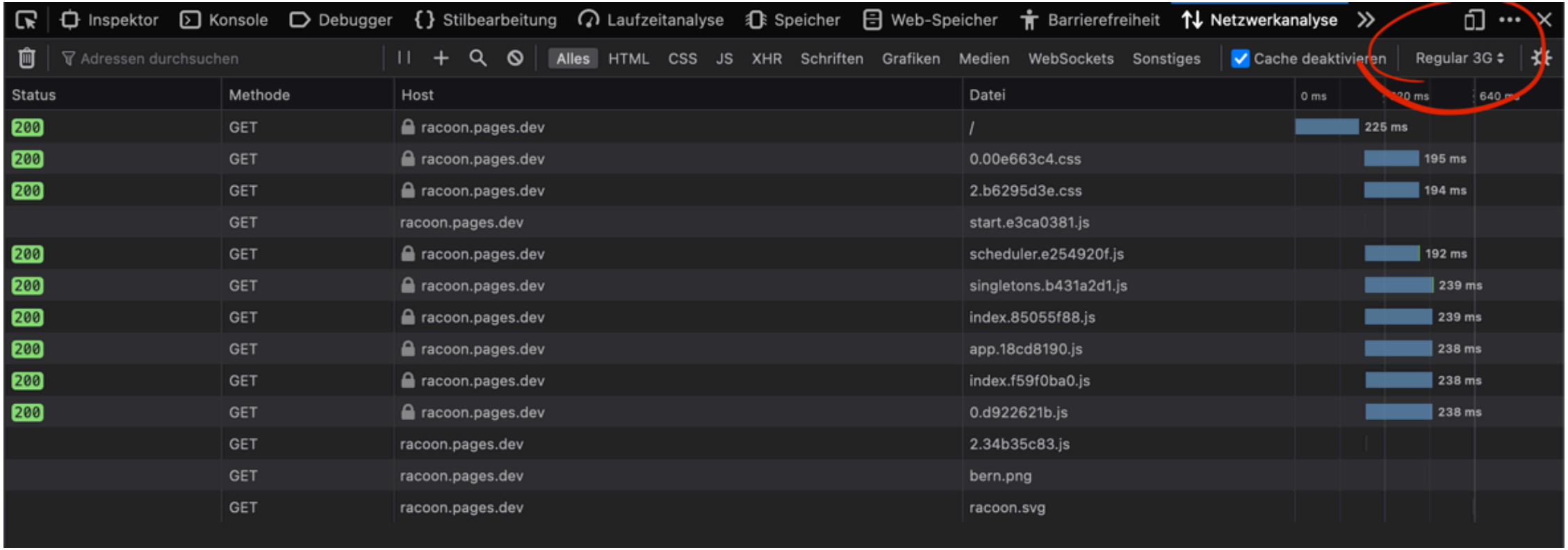
Energy, Time, Memory Comparision

Table 4: Normalized global results for Energy, Time, and Memory

Total					
	Energy (J)		Time (ms)		Mb
(c) C	1.00	(c) C	1.00	(c) Pascal	1.00
(c) Rust	1.03	(c) Rust	1.04	(c) Go	1.05
(c) C++	1.34	(c) C++	1.56	(c) C	1.17
(c) Ada	1.70	(c) Ada	1.85	(c) Fortran	1.24
(v) Java	1.98	(v) Java	1.89	(c) C++	1.34
(c) Pascal	2.14	(c) Chapel	2.14	(c) Ada	1.47
(c) Chapel	2.18	(c) Go	2.83	(c) Rust	1.54
(v) Lisp	2.27	(c) Pascal	3.02	(v) Lisp	1.92
(c) Ocaml	2.40	(c) Ocaml	3.09	(c) Haskell	2.45
(c) Fortran	2.52	(v) C#	3.14	(i) PHP	2.57
(c) Swift	2.79	(v) Lisp	3.40	(c) Swift	2.71
(c) Haskell	3.10	(c) Haskell	3.55	(i) Python	2.80
(v) C#	3.14	(c) Swift	4.20	(c) Ocaml	2.82
(c) Go	3.23	(c) Fortran	4.20	(v) C#	2.85
(i) Dart	3.83	(v) F#	6.30	(i) Hack	3.34
(v) F#	4.13	(i) JavaScript	6.52	(v) Racket	3.52
(i) JavaScript	4.45	(i) Dart	6.67	(i) Ruby	3.97
(v) Racket	7.91	(v) Racket	11.27	(c) Chapel	4.00
(i) TypeScript	21.50	(i) Hack	26.99	(v) F#	4.25
(i) Hack	24.02	(i) PHP	27.64	(i) JavaScript	4.59
(i) PHP	29.30	(v) Erlang	36.71	(i) TypeScript	4.69
(v) Erlang	42.23	(i) Jruby	43.44	(v) Java	6.01
(i) Lua	45.98	(i) TypeScript	46.20	(i) Perl	6.62
(i) Jruby	46.54	(i) Ruby	59.34	(i) Lua	6.72
(i) Ruby	69.91	(i) Perl	65.79	(v) Erlang	7.20
(i) Python	75.88	(i) Python	71.90	(i) Dart	8.64
(i) Perl	79.58	(i) Lua	82.91	(i) Jruby	19.84

https://haslab.github.io/SAFER/scp21.pdf

Browser Tools: Netzwerkanalyse



Browser Tools: Laufzeitanalyse